Locate and Label: Implementation of a Two-Stage Identifier for Nested Named Entity Recognition

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1. Problem Statement

Named Entity Recognition (NER) is a fundamental task in Natural Language Processing (NLP) that is traditionally treated as a sequence labeling problem. However, in specialized domains such as molecular biology, entities are frequently nested (e.g., a "DNA" sequence within a "Cell Type" mention).

Standard approaches using the BIO (Begin, Inside, Outside) tagging scheme fail because they assume each token belongs to exactly one entity. While "flat" NER models can achieve high accuracy on news datasets, they are mathematically incapable of representing the overlapping structures found in the GENIA corpus.

The core challenges addressed in this research are:

- **Overlapping Boundaries:** Tokens serving multiple semantic roles.
- **Boundary Sparsity:** The O(n^2) complexity of checking every possible span in a sentence, where most spans are irrelevant "background" noise.
- **Boundary Misalignment:** The difficulty of identifying exact start/end positions for complex biological terms.

2. Literature Survey

Nested Named Entity Recognition (NER) has been widely studied using several modeling approaches, including sequence-based, hypergraph-based, and span-based methods.

2.1 Sequence-based Methods

Sequence-based methods extend traditional NER models such as **BiLSTM-CRF (Finkel and Manning, 2009)**,[1] and **transformer-based taggers** to support nested entities. One of the earlier approaches in this direction is the **Layered-CRF model (Alex et al., 2007)**[6], which dynamically stacks multiple CRF layers to identify entities from inner to outer levels. In this approach, each layer performs flat NER, with its output serving as input to the next layer, enabling the model to progressively detect nested structures.

Building on this idea, more recent works use layered or pyramid structures **(Ju et al., 2018)**[2] to detect entities at multiple levels. By stacking multiple flat NER layers, the model can identify

nested mentions in a stepwise manner. However, the fundamental constraint of assigning a single label per token in each layer makes it difficult to represent complex overlapping entities efficiently and can lead to error propagation across layers.

2.2 Hypergraph-based Methods

Hypergraph-based methods model nested entities using **graph structures that represent multiple possible entity spans** simultaneously **(Lu and Roth, 2015[3]; Katiyar and Cardie, 2018)[4]**. These models capture complex relationships between entities and allow overlapping mentions to be represented within a single graph. Although effective at capturing structural dependencies, hypergraph-based models are often computationally expensive and complex to train due to the large state space of the graph transitions.

2.3 Span-based Methods

Span-based methods treat NER as a span classification task, where all possible spans in a sentence are generated and classified into entity categories (**Sohrab and Miwa, 2018**)[5]. These methods naturally support nested entities as each span is treated as an independent candidate. However, they often suffer from high computational costs due to the O(n^2) number of candidate spans. In addition, they may struggle to identify long entities and do not fully utilize boundary information, as they typically rely on fixed boundaries determined during the enumeration phase.

3. Dataset Description

The **GENIA Corpus** is a widely used benchmark dataset for biomedical named entity recognition (NER), particularly suited for evaluating models on **nested entity structures**. It consists of **2,000 MEDLINE abstracts** curated from the biomedical literature, primarily in molecular biology.

- **Categories:** The dataset covers five main types: **DNA, RNA, Protein, Cell Line, and Cell Type**.
- **Nesting Density:** Approximately **17-20%** of entities in the corpus are nested, making it significantly more complex than standard datasets like CoNLL-2003.
- **Train and test samples:** train.json contains approx. **16692 samples** and test.json contains **1854 samples**.
- **Pre-processing:**
    - Tokens are mapped to **Bio-word embeddings** (200-dimensional vectors trained on PubMed).
    - The data is split into Training, Validation (90-10 split of the train set), and a dedicated Test set.

4. Baseline Models

Two distinct architectures were implemented to compare performance.

4.1 Model 1: Span Classification Baseline

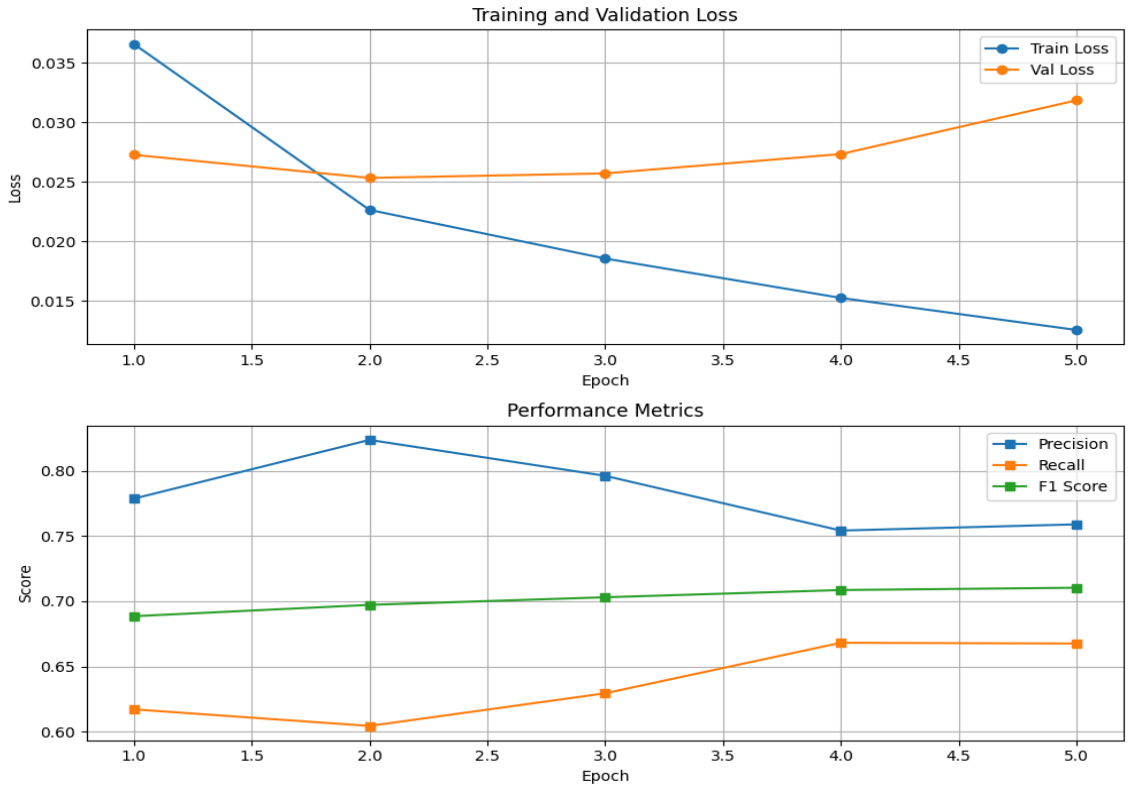
This baseline treats NER as an independent span classification task. Input tokens are encoded by a BiLSTM, and all candidate spans of length up to L are enumerated. Spans are represented by concatenating the hidden states of the start and end tokens, along with the mean-pooled representation of the internal tokens. An MLP classifier predicts entity types, including a `NONE` label. During inference, Hard-NMS is applied to resolve overlaps by retaining the highest-scoring spans.

4.2 Model 2: BiLSTM + layered CRF Tagging Baseline

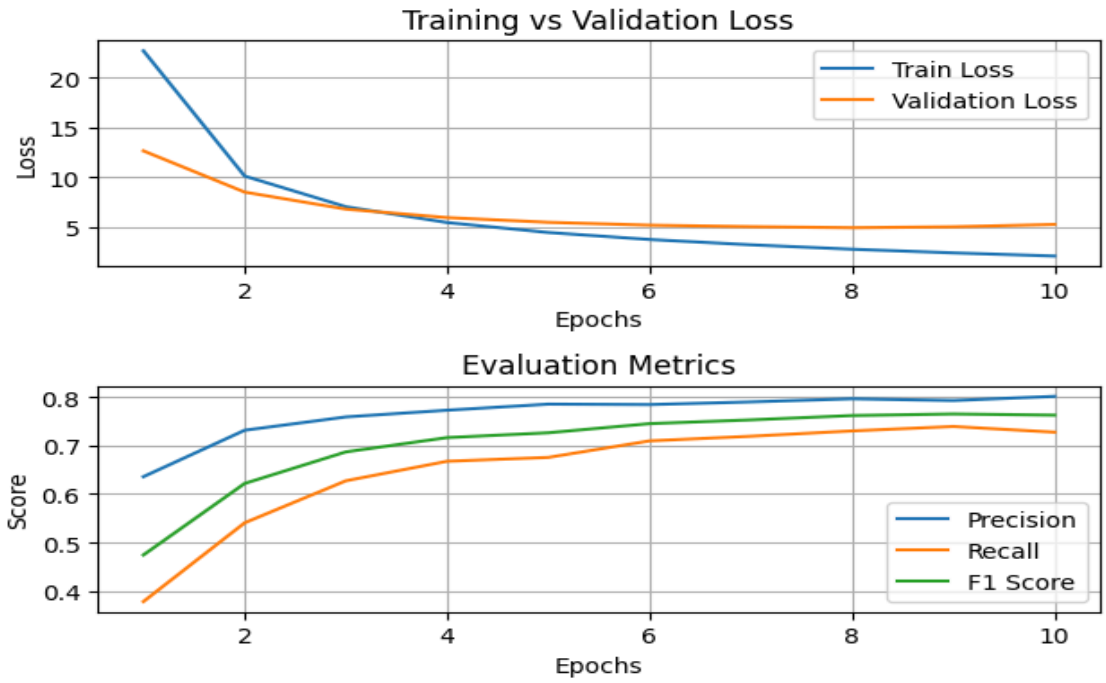
This model extends the standard BiLSTM-CRF architecture by stacking multiple CRF layers to approximate nested entity recognition. Tokens are encoded using a BiLSTM, and each CRF layer performs BIO tagging to identify entities at a particular level. The hidden representations from one layer are passed to the next, allowing the model to iteratively detect entities from inner to outer spans.

4.3 Training Performance Logs

Model 1: Span Classification Baseline



Model 2: BiLSTM + CRF Baseline



4.5 Test Set Results

The final evaluation on the hold-out test set reveals a clear performance gap between the two architectures:

Model	Checkpoint	Precision	Recall	F1-Score
Model 1 (Span)	baseline_1.pt	0.7117	0.5841	0.6416
Model 2 (CRF)	layered_crf_best.pt	0.7413	0.5804	0.6511

5. Model Architecture and Methodology

5.1. Locate and Label Architecture (Shen et al., ACL 2021)

The Locate and Label model formulates nested Named Entity Recognition (NER) as a two-stage span-based task inspired by object detection. It first enumerates candidate spans within a fixed window and encodes tokens using a BiLSTM-based encoder. Each span is represented using pooled token features and boundary representations.

In the first stage, a span proposal module filters candidate spans and predicts boundary offsets to refine their start and end positions. In the second stage, the refined spans are classified into entity types (including a background class). During inference, Soft Non-Maximum Suppression (Soft-NMS) is applied to handle overlapping predictions while preserving valid nested entities.

This design enables the model to capture nested structures effectively by separating localization and classification, while maintaining manageable computational complexity.

5.2. Ablation Study: Proposed Model (BioBERT + Attention Pooling)

As an ablation over the original architecture, we **replace the BiLSTM encoder with BioBERT** and introduce **attention-based span pooling**. BioBERT provides richer contextual representations, especially for biomedical text, while attention pooling enables the model to learn which tokens within a span are most informative, rather than relying on a single max-pooled feature.

The remaining architecture, including **span proposal, boundary regression, and classification**, is kept unchanged to isolate the effect of these modifications.

In terms of training dynamics, a clear difference is observed between the two models. The original architecture shows gradual, steady improvement over epochs and converges more slowly, requiring more training steps to reach peak performance. In contrast, the attention-based model learns significantly faster, with rapid gains in performance during the initial epochs and near-optimal performance within 5–6 epochs. This reflects more efficient representation learning due to the stronger BioBERT encoder and attention-based span pooling. Furthermore, this improved learning efficiency translates into better overall performance on the test set, with higher recall and F1-score. However, the faster convergence also introduces a slight reduction in precision and exact span accuracy, indicating a tendency to generate more false positives.

5.3. Evaluation Overview

The comparison between the original Locate and Label model (paper implementation) and the proposed attention-based ablation reveals clear differences in both training dynamics and final performance.

From the training logs, the attention-based model demonstrates significantly faster convergence. It achieves strong performance within the first few epochs, reaching an F1-score above 0.80 by around epoch 5–6. In contrast, the original model improves more gradually, requiring many more epochs to reach similar performance levels. This indicates that the BioBERT encoder and attention-based span pooling enable more efficient feature learning early in training.

Despite the difference in convergence speed, both models ultimately reach comparable peak performance. The paper model achieves a best development F1-score of **0.8208**, while the ablation model slightly surpasses it with a best F1-score of **0.8218**. This suggests that while the

architectural improvements primarily impact learning efficiency, they also provide a small but consistent gain in final performance.



A more detailed analysis shows a trade-off between precision and recall. The ablation model tends to achieve higher recall across epochs, indicating improved ability to detect entities, particularly in complex or nested scenarios. However, this is often accompanied by slightly lower precision, reflecting an increase in false positives. The original model, on the other hand, maintains more stable precision while improving recall more conservatively.

In terms of Exact Match (EM), both models show similar trends, with gradual improvements as training progresses. The differences between the two are relatively small, indicating that both approaches are capable of learning accurate span boundaries, although the ablation model occasionally sacrifices boundary precision for higher recall.

Overall, the results demonstrate that the proposed ablation improves training efficiency and slightly enhances overall performance, while introducing a minor trade-off in precision. The use of BioBERT and attention-based pooling leads to faster convergence and better generalization, making it a more effective variant of the original Locate and Label framework..

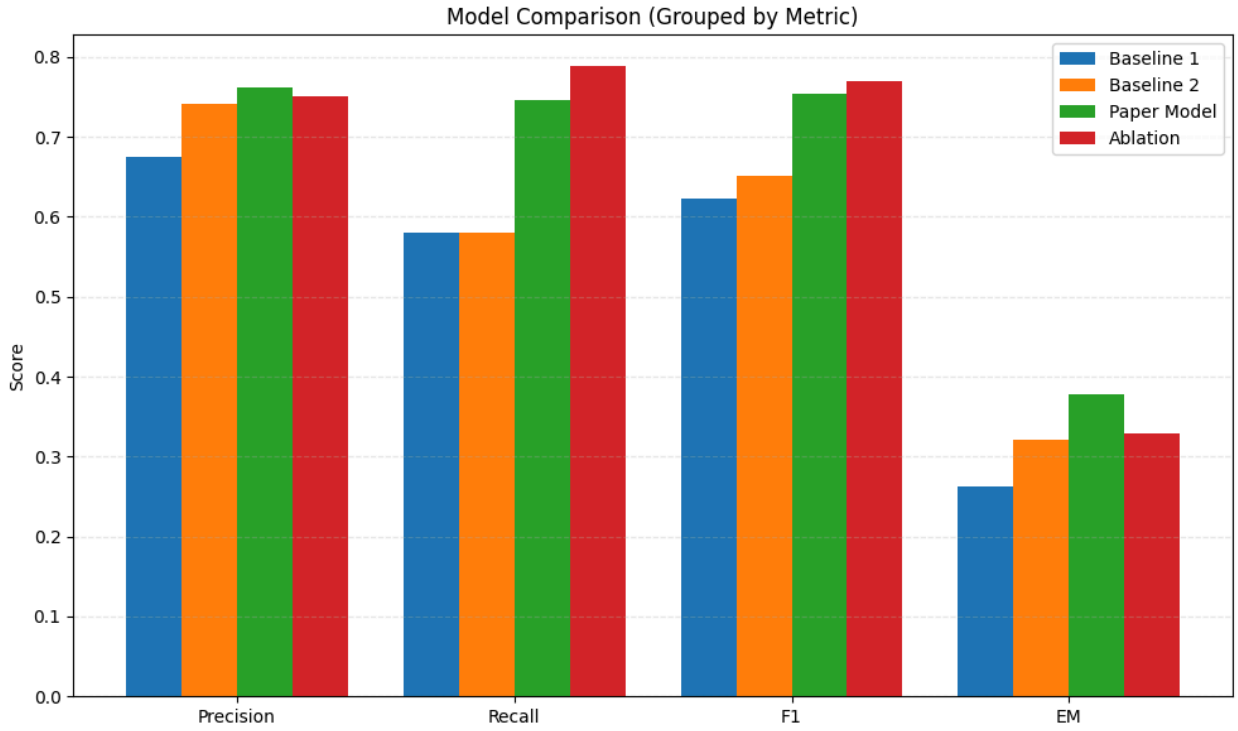
The table below shows the results over the test dataset.

Model	**Precision**	**Recall**	**F1-Score**	**EM**
Model	0.7620	0.7459	0.7539	0.3776

Ablation	0.7510	0.7882	0.7692	0.3288

6. Conclusion

This work evaluates the Locate and Label framework for nested NER and investigates the impact of incorporating transformer-based representations and attention-based span pooling. The original architecture provides strong and stable performance, achieving higher precision and better exact span matching, but requires more epochs to converge.



The proposed ablation with BioBERT and attention pooling significantly improves training efficiency, converging rapidly within the first few epochs. More importantly, it achieves a higher overall F1-score on the test set, driven by improved recall and better entity detection, particularly in complex or nested cases.

However, this improvement comes at the cost of slightly reduced precision and lower Exact Match scores, indicating an increase in false positives and less accurate boundary localization. These results highlight a trade-off between detection capability and prediction precision.

In conclusion, the attention-based model offers a more effective and efficient approach to nested NER, achieving better overall performance, but requires further refinement to control false positives and improve boundary accuracy. Future work can focus on improving span filtering and calibration, or on incorporating stricter boundary constraints to better balance recall and precision.

7. References

1. Jenny Rose Finkel and Christopher D. Manning. 2009. Nested Named Entity Recognition. In *Proceedings of the 2009 Conference on Empirical Methods in Natural Language Processing*, pages 141–150, Singapore. Association for Computational Linguistics.
2. Meizhi Ju, Makoto Miwa, and Sophia Ananiadou. 2018. A Neural Layered Model for Nested Named Entity Recognition. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1446–1459, New Orleans, Louisiana. Association for Computational Linguistics.
3. Wei Lu and Dan Roth. 2015. Joint Mention Extraction and Classification with Mention Hypergraphs. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 857–867, Lisbon, Portugal. Association for Computational Linguistics.
4. Arzoo Katiyar and Claire Cardie. 2018. Nested Named Entity Recognition Revisited. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 861–871, New Orleans, Louisiana. Association for Computational Linguistics.
5. Mohammad Golam Sohrab and Makoto Miwa. 2018. Deep Exhaustive Model for Nested Named Entity Recognition. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2843–2849, Brussels, Belgium. Association for Computational Linguistics.
6. Beatrice Alex, Barry Haddow, and Claire Grover. 2007. Recognising nested named entities in biomedical text. In Biological, translational, and clinical language processing, pages 65–72, Prague, Czech Republic. Association for Computational Linguistics.